Summary in Graph

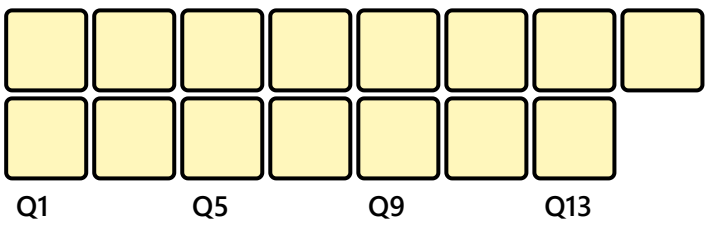
Exam Summary (GO Classes CS/DA 2026| Linear Algebra | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (213) out of 2125 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Linear Algebra

Let $\left\{\overrightarrow{\mathbf{u}_1}, \overrightarrow{\mathbf{u}_2}, \overrightarrow{\mathbf{u}_3}\right\}$ be a linearly independent set in $\mathbb{R}^3$. Then, for a scalar c, what can be concluded about the set $\left\{c\overrightarrow{\mathbf{u}_1}, c\overrightarrow{\mathbf{u}_2}, c\overrightarrow{\mathbf{u}_3}\right\}$?

A. The set $\left\{c\overrightarrow{\mathbf{u}_1}, c\overrightarrow{\mathbf{u}_2}, c\overrightarrow{\mathbf{u}_3}\right\}$ is linearly independent for all values of $c \neq 0$

B. The set $\left\{c\overrightarrow{\mathbf{u}_1}, c\overrightarrow{\mathbf{u}_2}, c\overrightarrow{\mathbf{u}_3}\right\}$ is linearly dependent for all values of $c \neq 0$

C. The set $\left\{c\overrightarrow{\mathbf{u}_1}, c\overrightarrow{\mathbf{u}_2}, c\overrightarrow{\mathbf{u}_3}\right\}$ is linearly independent only if $c = 1$

D. The set $\left\{c\overrightarrow{\mathbf{u}_1}, c\overrightarrow{\mathbf{u}_2}, c\overrightarrow{\mathbf{u}_3}\right\}$ is linearly dependent only if $c = 0$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

What is the definition of linear independence for a set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$? Let $x_1, x_2, \ldots, x_m$ be scalars.

A. $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the only solution to $x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_m\mathbf{u}_m = \mathbf{0}$ is $x_1 = x_2 = \cdots = x_m = 0$.

B. $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if there exists a non-trivial solution to $x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_m\mathbf{u}_m = \mathbf{0}$.

C. $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the vectors are orthogonal to each other.

D. $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the vectors are not scalar multiples of each other.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 03sec | Discuss

Q #3

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let us consider the following three vectors v_1, v_2, and v_3 in $\mathbb{R}^3$.

$$\begin{aligned} v_1 &= \begin{pmatrix} 1 & 0 & -2 \end{pmatrix} \\ v_2 &= \begin{pmatrix} -1 & 0 & 2 \end{pmatrix} \\ v_3 &= \begin{pmatrix} 1 & 0 & 2 \end{pmatrix}. \end{aligned}$$

A. The three vectors v_1, v_2, and v_3 are linearly independent.

B. Every pair of vectors $\{v_1, v_2\}$, $\{v_2, v_3\}$, and $\{v_1, v_3\}$ are linearly independent.

C. All pairs except the pair $\{v_1, v_2\}$ are linearly independent.

D. All pairs except the pair $\{v_1, v_3\}$ are linearly independent.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let a, b be in $\mathbb{R}$. Consider the three vectors

$$\boldsymbol{v}_1 = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 0 \\ b \\ 1 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

For which values of a and b are $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ independent?

A. $a = 0$ and $b = 1$

B. $a \neq 0$ and $b \neq 1$

C. $a = 0$ and $b \neq 1$

D. $a \neq 0$ and $b = 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Which of the following is FALSE?

A. If $v_1, \ldots, v_4$ are in $\mathbf{R}^5$ and $\{v_1, v_2, v_3\}$ is linearly dependent then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.
B. If $v_1, \ldots, v_4$ are in $\mathbf{R}^5$ and v_1 is not a linear combination of $\{v_2, v_3, v_4\}$, then $\{v_1, v_2, v_3, v_4\}$ is linearly independent.
C. If $v_1, \ldots, v_4$ are linearly independent vectors in $\mathbf{R}^5$, then $\{v_1, v_2, v_3\}$ is also linearly independent.
D. Any set of 6 vectors in $\mathbf{R}^5$ is linearly dependent.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider a set of n linearly independent vectors $\{\vec{w}_1, \ldots, \vec{w}_n\} \in \mathbb{R}^n$. A vector $\vec{u} \in \mathbb{R}^n$ will:

- Option 1. Always be a linear combination of $\{\vec{w}_1, \ldots, \vec{w}_n\}$
- Option 2. Never be a linear combination of $\{\vec{w}_1, \ldots, \vec{w}_n\}$

Enter the correct option as the numeric number. That is if option 2 is correct then enter 2.

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following set is/are linearly independent?

A. $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$

B. $\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$

C. $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

D. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

$$a\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}+b\begin{bmatrix}-1\\2\\-3\\4\\-5\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\\0\end{bmatrix}$$

How many number of pairs (a, b) are there, that satisfy the above equation?

A. 0
B. 1
C. Infinite
D. 2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Given a set of vectors $\mathrm{S}(|\mathrm{S}| \geq n)$, with all vectors in $\mathbb{R}^n$, which of the following is a necessary and sufficient condition for the vectors of S to be Linearly Dependent?

A. Exactly n vectors can be represented as a linear combination of other vectors of the set S.
B. At least n vectors can be represented as a linear combination of other vectors of the set S.
C. At least one vector u can be represented as a linear combination of any vector(s) of the set S.
D. At least one vector u can be represented as a linear combination of vectors (other than u) of the set S.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is(are) sufficient argument(s) to show that the vectors of set S are linearly dependent?

$$\mathrm{S}=\left\{u=\begin{bmatrix}1\\-2\\7\end{bmatrix}, v=\begin{bmatrix}-7\\14\\-49\end{bmatrix}, w=\begin{bmatrix}0\\0\\0\end{bmatrix}\right\}$$

Treat each option independently, correct option independently should be sufficient to infer that vectors are linearly dependent.

A. $0u+0v+1w=\begin{bmatrix}0\\0\\0\end{bmatrix}$

B. $0u+0v=w$

C. $0u+0v+0w=\begin{bmatrix}0\\0\\0\end{bmatrix}$

D. $7u+(-1)v+1w=\begin{bmatrix}0\\0\\0\end{bmatrix}$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE?

A. If $v_1, \ldots, v_4$ are in $\mathbb{R}^5$ and $\{v_1, v_2, v_3\}$ is linearly dependent, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.
B. If $v_1, \ldots, v_4$ are in $\mathbb{R}^5$ and v_1 is not a linear combination of $\{v_2, v_3, v_4\}$, then $\{v_1, v_2, v_3, v_4\}$ is linearly independent.
C. If $v_1, \ldots, v_4$ are linearly independent vectors in $\mathbb{R}^5$, then $\{v_1, v_2, v_3\}$ is also linearly independent.
D. All of the above.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is(are) true for the following system of linear equations $\mathrm{AX} = \overrightarrow{0}$

$$\begin{bmatrix} 2 & 3 & -5 \\ -5 & -1 & 32 \\ 2 & -4 & -26 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

A. $\mathrm{X} = [-7 \quad 3 \quad 1]^\top$ is a solution to the equation $\mathrm{AX} = \overrightarrow{0}$.
B. Showing that $\mathrm{X} = [0 \quad 0 \quad 0]^\top$ is a solution to $\mathrm{AX} = \overrightarrow{0}$ is sufficient to conclude that the columns of A are Linearly Independent.
C. Showing that $\mathrm{X} = [-7 \quad 3 \quad 1]^\top$ is a solution to $\mathrm{AX} = \overrightarrow{0}$ is sufficient to conclude that the columns of A are Linearly Independent.
D. $\mathrm{X} = [0 \quad 0 \quad 0]^\top$ is a solution to the equation $\mathrm{AX} = \overrightarrow{0}$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Given a set S of vectors in $\mathbb{R}^n$, and set $\mathrm{X}(\mathrm{X} \subset \mathrm{S})$ and $\mathrm{Y}(\mathrm{S} \subset \mathrm{Y})$, mark all the statements which are always true:

A. If the vectors of S are Linearly Dependent, then the vectors of X are also Linearly Dependent.
B. If the vectors of S are Linearly Independent, then the vectors of X are also Linearly Independent.
C. If the vectors of S are Linearly Dependent, then the vectors of Y are also Linearly Dependent.
D. If the vectors of S are Linearly Independent, then the vectors of Y are also Linearly Independent.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are CORRECT?
$u, v,$ and w are vectors in $\mathbb{R}^n$.

A. A set $\{u, v, w\}$ is linearly independent if u can not be written as linear combination of v and w.
B. A set $\{u, v, w\}$ is linearly dependent if u is a linear combination of v and w.
C. If a set $\{u, v, w\}$ is linearly dependent then u is a linear combination of v and w.
D. For three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, if $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent, and $\{\mathbf{v}, \mathbf{w}\}$ is linearly independent; then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

- $\mathrm{S1}$: A set of two vectors in $\mathbb{R}^n$ is Linealy dependent if at least one vector is a multiple of the other.
- $\mathrm{S2}$: A set of n vectors in $\mathbb{R}^n$ is Linealy independent if and only if none of the vectors are a multiple of any other vector.

A. $\mathrm{S1}$ and $\mathrm{S2}$ both are correct
B. $\mathrm{S1}$ is correct and $\mathrm{S2}$ is incorrect
C. $\mathrm{S2}$ is correct and $\mathrm{S1}$ is incorrect
D. $\mathrm{S1}$ and $\mathrm{S2}$ both are incorrect

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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